



WISCONSIN
UNIVERSITY OF WISCONSIN-MADISON

Name: _____

Math 421 - Section 2 - Exam 1

Fall Semester 2024

October 10, 2024 - 1:00 pm to 2:15 pm

Instructions:

- Read each problem carefully.
- Write legibly.
- Show all your work on these sheets. Feel free to use the opposite side.
- This exam has 6 pages, and 5 problems. Please make sure that all pages are included.
- You may not use books, notes, calculators, etc.

Good luck!

Prob.	Points possible	Score
1	6	
2	4	
3	10	
4	10	
5	10	
6	10	
TOTAL	50	

Question 1 (TRUE or FALSE). (6 points, 1 points each) Determine if each statement is true or false. You do not need to justify your answer.

1. ____ The statement “If P , then Q ” is logically equivalent to the statement “If (not P), then (not Q).”
2. ____ The statement “not(P and Q)” is logically equivalent to the statement “(not P) and (not Q).”
3. ____ If a is irrational and b is rational, then $a \cdot b$ is irrational.
4. ____ If a is irrational and b is irrational, then $a + b$ is irrational.
5. ____ If (a_n) converges, then (a_n) is bounded.
6. ____ If (a_n) is bounded and monotone, then (a_n) converges.

Question 2. (4 points) Determine the minimum, maximum, infimum, and supremum of the following set. You do not need to justify your answer.

$$A = \left\{ 1 - \frac{1}{n} : n \in \mathbb{N} \right\}.$$

Minimum: _____

Maximum: _____

Infimum: _____

Supremum: _____

Question 3. *(Total of 10 points) Prove: If a is rational and b is irrational, then $a + b$ is irrational.*

Question 4. (Total of 10 points) Let (a_n) be the sequence where $a_1 = 1$, $a_2 = 2$, and

$$a_n = a_{n-1} + 3a_{n-2} \quad \text{when } n \geq 3.$$

Prove that $a_n \leq 3^n$ for all $n \geq 1$.

Question 5. *(Total of 10 points) Prove, using only the definition, that $\lim_{n \rightarrow \infty} \frac{n}{4n+1} = \frac{1}{4}$.*

Question 6. (Total of 10 points) Prove, using only the definition, that: if (a_n) is a sequence where $a_n \geq 0$ for all n and $\lim a_n = a$, then

$$\lim \frac{1}{7 + a_n} = \frac{1}{7 + a}.$$